Little Lab Coats

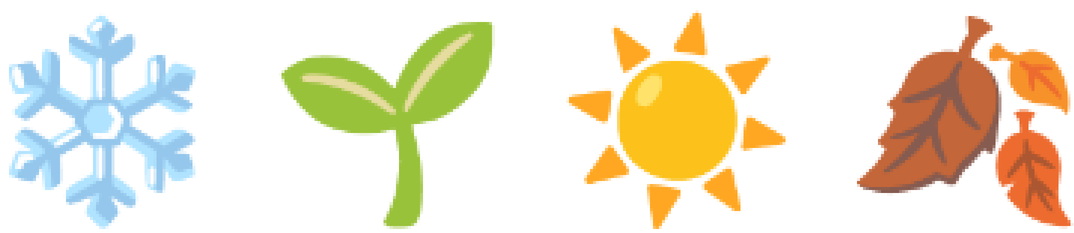
📅 How Much Daylight? Seasons and the Sun

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 4 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars
🏠 **Materials:** Household only

⚠️ Sun Safety Reminder

This lesson involves observing daylight patterns — not looking at the sun. As always, **never look directly at the sun.** Daylight observations mean noticing when it's light or dark outside — keep eyes away from the sun itself.



Winter, Spring, Summer, Fall — daylight hours change dramatically across the seasons

📋 **NGSS Standard: 1-ESS1-2** — Make observations at different times of year to relate the amount of daylight to the time of year.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Compare* the hours of daylight across the four seasons
- *Explain* the pattern: more daylight in summer, less in winter
- *Identify* Earth's tilt (not distance from the sun) as the cause of seasons
- *Create* a bar graph showing seasonal daylight changes

🔬 The Big Science Idea

Have you ever noticed that summer days last longer and winter days seem short? This is real! The number of daylight hours changes throughout the year in a predictable pattern.

In summer, the sun rises early, climbs high in the sky, and sets late — giving us more hours of light. In winter, the sun rises later, stays lower in the sky, and sets early — giving us fewer hours of light.

Here's the surprising part: this has *nothing* to do with how close Earth is to the sun! Earth is actually slightly *closer* to the sun in January (winter in the Northern Hemisphere) than in July (summer). The cause of seasons is Earth's *tilt*.

Earth spins at a tilt of about 23.5 degrees. In summer, our hemisphere is tilted *toward* the sun — days are longer and sunlight hits us more directly (more energy, more warmth). In winter, our hemisphere is tilted *away* — days are shorter and sunlight hits us at a glancing angle (less energy, less warmth).

> 💡 **Key vocabulary:** season, daylight, sunrise, sunset, tilt, hemisphere, pattern, summer solstice, winter solstice, equinox

> ☀️ **Sample Daylight Hours (Denver, CO — approximate):**
> Winter solstice (Dec 21): ~9 hours 21 minutes of daylight
> Spring equinox (Mar 20): ~12 hours 10 minutes of daylight
> Summer solstice (Jun 21): ~14 hours 58 minutes of daylight
> Fall equinox (Sep 22): ~12 hours 10 minutes of daylight
> *Look up your location's actual times at **timeanddate.com/sun***

📦 Materials

🏠 What You'll Need

- Paper and crayons
- Ruler (to draw the bar graph)
- A flashlight or lamp
- A ball, orange, or small globe to act as Earth
- 4 strips of paper (or sentence strips) for comparing daylight lengths
- Sunrise/sunset data — parent looks up for your location at *timeanddate.com/sun* or a weather app
- This worksheet (printed)

🔬 The Daylight Data Activity

Step 1 — Act Out the Seasons
Start with a quick movement model. Put a lamp or flashlight in the middle of the room as the sun. Have your child hold a ball, orange, or small globe as Earth and keep it tilted the same way the whole time. Slowly walk around the light and stop at "summer" and "winter." Ask: *"Is our side getting more light or less light now?"* Keep this very simple. The goal is just to notice that the tilt changes how much light our side gets.

Step 2 — Try a Flashlight Demo
Shine the flashlight at the tilted ball. Show how the light hits more directly in summer and at more of a slant in winter. Let your child turn the ball and point to the part getting the most light. Then connect it back to daylight: when our side is tilted toward the sun, we get longer days.

Step 3 — Gather the Data
Parent: Before or during the lesson, look up the sunrise and sunset times for your location for four dates:

- Around **December 21** (winter solstice — shortest day)
- Around **March 20** (spring equinox — equal day/night)
- Around **June 21** (summer solstice — longest day)
- Around **September 22** (fall equinox — equal day/night)

Write the sunrise and sunset times in the Data Table on the worksheet.

Step 4 — Build Daylight Strips and Graph the Pattern
First, make four paper strips for winter, spring, summer, and fall. Cut or compare the strips so winter is shortest, summer is longest, and spring and fall are in between. Line them up from shortest to longest. Then use the same information to color in the bar graph on the worksheet. Use different colors for each season.

Step 5 — Read the Pattern
Look at the completed strips and graph together. Ask: *"Which season has the most daylight? Which has the least? What pattern do you see as the year goes from winter to summer and back again?"*

Step 6 — Check Real Life Outside
If possible, step outside and notice where shadows are or talk about a time earlier in the day when shadows looked different. Connect the idea back to the lesson: the sun's path changes across the year, and that helps create different amounts of daylight in different seasons.

💬 Discussion Questions

🗣️ **Talk about these together:**

- Which season did you enjoy having more daylight in? Why?
- How would your day be different if the sun set at 4pm vs. 9pm?
- People often say "in summer we're closer to the sun." Do you think that's true? (Actually, we're closer to the sun in winter!)
- Can you think of animals that might behave differently based on how much daylight there is? (Bears hibernating, birds migrating!)

🌟 **Big Question: "In Alaska in summer, the sun doesn't set for weeks. How can people sleep?"**

This is called the Midnight Sun and it happens above the Arctic Circle. The tilt is so extreme that the sun stays above the horizon 24 hours a day for weeks. People use blackout curtains, eye masks, and adjust their sleep schedules. In winter, the opposite happens — the sun doesn't rise for weeks (Polar Night). Residents report it affects mood and energy levels significantly, a condition called Seasonal Affective Disorder (SAD).

🚀 Extensions

- 🌍 **Flip the Seasons:** When it's summer in the Northern Hemisphere, it's winter in Australia (Southern Hemisphere). Use the flashlight and ball again, but point to the opposite side of Earth to show why.
- 📏 **Daylight Strip Match:** Make new paper strips for different months and have your child put them in order from shortest daylight to longest daylight.
- 👣 **Shadow Walk:** Go outside in the morning and later in the afternoon and compare shadow length and direction. Talk about how the sun's position changes through the day.
- 🌱 **Plant and Animal Connection:** Talk about animals, flowers, or bedtime routines that change when days are longer or shorter.

👩‍🏫 Parent/Teacher Notes

- **Seasons ≠ distance from sun:** This is one of the most common misconceptions in science. Earth is slightly closer to the sun in January. The seasons are caused by tilt, not distance. This is worth spending extra time on.
- **The "solstice" words:** "Solstice" means "sun stands still" in Latin — at the solstices, the sun's position at noon stops getting higher or lower for a few days. "Equinox" means "equal night" — day and night are roughly equal in length.
- **Hemisphere specifics:** This lesson describes the Northern Hemisphere (where most US families live). If you're in the Southern Hemisphere, the seasons are reversed — December is summer, June is winter.
- **Grade 1 depth:** For this age, the key takeaways are: (1) daylight hours change through the year, (2) summer has more, winter has less, (3) this creates a predictable repeating pattern. Keep the tilt explanation very concrete with the flashlight and ball model.
- **Make it hands-on:** The strongest version of this lesson is movement first, graph second. Let the child hold the "Earth," walk around the "sun," and build paper daylight strips before doing the worksheet.
- **The bar graph:** Grade 1 students may need help drawing the bars accurately. It's fine for the parent to draw the axes and label them while the child colors in the bars and observes the pattern.

📅 Student Worksheet — Daylight Bar Graph

Unit 1, Lesson 4 — Daylight and Seasons

Name: ________________________________ Date: ________________

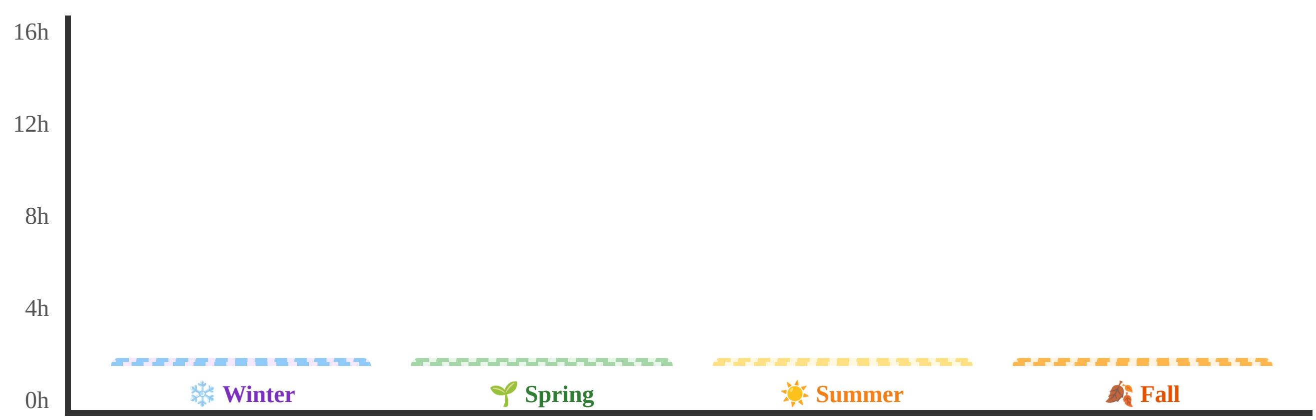
📊 Daylight Data Table

Fill in the sunrise, sunset, and total daylight hours for each season. Your parent will help you find the data!

Season	Date Used	Sunrise Time	Sunset Time	Total Hours of Daylight
❄️ **Winter**	Dec 21			
🌱 **Spring**	Mar 20			
☀️ **Summer**	Jun 21			
🍂 **Fall**	Sep 22			

📊 Color the Bar Graph

Color each bar up to the number of daylight hours for that season. Use a different color for each season! The Y-axis goes from 0 to 16 hours.



Color each bar from the bottom up to show the hours of daylight for that season!

✏️ Answer These Questions

1. Which season has the *most* hours of daylight? ______________________
2. Which season has the *fewest* hours of daylight? ______________________

3. Do the hours of daylight follow a pattern? (circle) **YES** / **NO**

4. What causes the seasons? (circle the correct answer):

- ☐ Earth gets closer to the sun in summer
- ☐ Earth tilts toward or away from the sun
- ☐ The sun gets bigger in summer

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

📊 Sample Daylight Data (Denver, CO)

- Winter (Dec 21): Sunrise ~7:17am, Sunset ~4:38pm → **~9 hours 21 min**
- Spring (Mar 20): Sunrise ~6:21am, Sunset ~6:31pm → **~12 hours 10 min**
- Summer (Jun 21): Sunrise ~5:31am, Sunset ~8:29pm → **~14 hours 58 min**
- Fall (Sep 22): Sunrise ~6:38am, Sunset ~6:48pm → **~12 hours 10 min**

Look up your specific location at timeanddate.com/sun for accurate local times.

✏️ Worksheet Answers

1. **Summer** has the most hours of daylight (~15 hours at the summer solstice in Denver).
2. **Winter** has the fewest hours of daylight (~9 hours at the winter solstice in Denver).
3. **Yes** — the hours follow a predictable pattern that repeats every year.
4. The correct answer is: *Earth tilts toward or away from the sun.* Earth's tilt causes seasons — not its distance from the sun.

📈 What the Graph Should Show

The bar graph should show: Winter (shortest bar) → Spring (medium-tall) → Summer (tallest bar) → Fall (medium-tall, similar to Spring). The pattern looks like a hill: it rises from winter to summer, then falls again toward winter.

✅ What Success Looks Like

- Child can read the bar graph and identify which season has more or less daylight
- Child can describe the seasonal pattern (more daylight in summer, less in winter)
- Child understands this pattern repeats every year (predictable)
- Bonus: Child can explain that Earth's tilt causes seasons, not closeness to the sun